

Batched pattern search for QAOA parameter optimization on cloud-accessed quantum hardware

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On a cloud-accessed quantum processor the classical optimizer of a variational algorithm communicates with the device through submitted jobs, and each job carries a queueing and turnaround overhead that does not depend on how many circuits it contains. We study a simple consequence of this for the quantum approximate optimization algorithm (QAOA). An optimizer whose next set of trial parameters is known before any result returns can evaluate the whole set in one job, whereas a sequential optimizer such as COBYLA spends one job per function evaluation. We compare a batched pattern search with COBYLA on IBM's `ibm_fez` processor for cardinality-constrained portfolio selection with six to twelve assets, giving both optimizers the same number of jobs and the same number of shots. At every size the batched search reaches, after its first job, a parameter quality that the serial optimizer takes several jobs to match, and the two methods converge to comparable final values; repeating the eight-asset comparison from four starting points gives the same ordering each time. The instances are small enough to be solved exactly, which lets us check that the device's output distribution is correlated with the true ranking of portfolios and is clearly separated from a control circuit of the same depth with the cost layer removed. A short scan over circuit depth at classically optimal parameters shows that on this device the measured solution quality peaks at two or three QAOA layers. The protocol is described in enough detail to be reproduced, and the notebooks and raw counts are released.

I. INTRODUCTION

A variational quantum algorithm consists of a classical optimizer wrapped around a parameterized quantum circuit [1–3], and QAOA in particular has been run on superconducting processors at a range of sizes [4, 5]. When the circuit runs on a processor reached through a cloud queue, every exchange between the optimizer and the device is a submitted job, and a job has costs that do not depend on what it contains: it waits in a queue, it is loaded onto the control electronics, and its results travel back. In the runs reported here the shortest round trip we observed on IBM's `ibm_fez` was about 11s and the median about 19s, and the shortest did not depend on what the job carried. An optimizer that submits one circuit evaluation per job pays this overhead once per evaluation. Queueing and turnaround on shared quantum hardware have been characterized before [6–8], and their consequences for stochastic optimizers have been analyzed theoretically [9]. Our aim here is narrower and more practical: to compare two classical optimizers for QAOA on real hardware when the number of submitted jobs, rather than the number of circuit evaluations, is the quantity held fixed.

The comparison rests on a simple observation. A single job may carry many circuits — in Qiskit's primitive interface one primitive unified bloc (PUB) can hold an arbitrary number of parameter vectors for the same circuit [10] — but an optimizer can use that capacity only to the extent that it can name its next trial points before

it sees any result. Methods differ in how many points that is. COBYLA [11], Nelder–Mead [12] and Powell's method [13] are sequential once past their initial simplex or direction set, since each new point depends on the value just returned. SPSA [14] names two points per iteration, and a parameter-shift gradient [15, 16] names all of its evaluations at once. Pattern search [17–19] names its whole iteration in advance: the current point, a fixed stencil of trial points around it, and, in the variant used here, one probe along the last accepted step. For a QAOA circuit with p layers that is $4p + 2$ points, and all of them can be submitted as the rows of one job.

We compare this batched pattern search with COBYLA on four instances of cardinality-constrained portfolio selection with 6, 8, 10 and 12 assets, executed on `ibm_fez`, with both optimizers given the same number of jobs and the same total number of shots. COBYLA is used as the serial reference because it was the strongest of the common sequential methods in the simulations we ran before using any hardware time (Appendix A). The instances are small enough that the exact optimum and the exact ranking of every feasible portfolio are known by enumeration, so the quality of any parameter vector can be evaluated exactly and the device output can be compared with the true ordering. Two controls accompany each run: a circuit of identical depth with the cost layer switched off, which measures the noise floor, and a repeat of the first job at the end of the run, which bounds drift. Two smaller follow-up experiments — a scan over circuit depth at classically optimal parameters, and a repeat of the 8-asset comparison from different starting points — address questions raised by the main runs.

The results are what the counting argument leads one to expect, and their value, such as it is, lies in having been

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measured on hardware with the controls in place. The batched search reaches a good region of parameter space in one or two jobs where COBYLA needs several; the two methods reach comparable final quality; and the device output at the returned parameters is correlated with the exact ranking of portfolios and well separated from the control. We also find that on this device the measured solution quality peaks at two or three QAOA layers and declines beyond that, which is useful to know when choosing the depth. None of this constitutes an advantage over classical solvers, which handle these instances instantly; the sizes were chosen so that every claim could be checked against an exact answer, not for difficulty.

The paper is written so that the experiment can be followed and repeated. Section II states the optimization problem and its Ising form, Sec. III constructs the circuit, Sec. IV describes what a job costs and how the runs were budgeted, Sec. V states the optimizer, Sec. VI fixes the protocol and its controls, Sec. VII reports the hardware results, and Sec. VIII discusses their limitations. Appendix A records how the hyperparameters were chosen in simulation and Appendix B describes the released material and some implementation pitfalls.

II. THE PROBLEM AND ITS ISING FORM

We solve the discrete, cardinality-constrained form of Markowitz mean–variance portfolio selection [20], a common test problem for quantum optimization [21–24]. Let $x_i \in \{0, 1\}$ indicate that asset i is held, let $\mathbf{r}$ be the vector of expected returns and Σ the covariance matrix. With a hard budget of exactly B assets, the cost function is

$$C(\mathbf{x}) = \lambda \mathbf{x}^T \Sigma \mathbf{x} - \mathbf{r}^T \mathbf{x} + \mu \left(\sum_i x_i - B \right)^2, \quad (1)$$

where the last term is the usual quadratic penalty that turns the constraint into a QUBO [25, 26]. Two coefficients in Eq. (1) affect whether the problem is meaningful, and we describe how each was set.

The risk aversion λ . If λ is small the quadratic term is irrelevant and the problem reduces to sorting assets by expected return. We fix λ so that, over the feasible set, the range of the risk term equals the range of the return term,

$$\lambda = \frac{\max_{\text{feas}}(\mathbf{r}^T \mathbf{x}) - \min_{\text{feas}}(\mathbf{r}^T \mathbf{x})}{\max_{\text{feas}}(\mathbf{x}^T \Sigma \mathbf{x}) - \min_{\text{feas}}(\mathbf{x}^T \Sigma \mathbf{x})}, \quad (2)$$

so that the optimum is a trade-off between the two terms rather than a ranking by one of them.

The penalty μ . If μ is too small the global minimum of Eq. (1) is infeasible; if it is too large the penalty dominates every Hamiltonian coefficient, the ZZ couplings become nearly uniform, and the ansatz spends its layers enforcing the Hamming weight rather than discriminating among feasible portfolios. We scanned μ against the exactly optimized noiseless $p = 2$ QAOA (Appendix A)

TABLE I. The four nested instances. $r(\text{uniform})$ is the expected approximation ratio of the uniform distribution over the feasible set, which any method must exceed to have done anything. All quantities are exact, from enumeration of all 2^n bitstrings.

n	B	λ	μ	$\binom{n}{B}$	$C_{\min}$	$C_{\max}^{\text{feas}}$	$r(\text{uniform})$
6	3	1.771	8	20	−136.8	21.1	0.451
8	4	1.653	9	70	−161.1	23.3	0.456
10	5	1.553	12	252	−219.4	23.7	0.490
12	6	1.360	13	924	−243.3	26.0	0.505

and took the value at which the noiseless approximation ratio was largest, which is about 5% of the spread of the penalty-free objective over the feasible set. At all four sizes this is comfortably above the smallest penalty that makes the ground state feasible.

Under $x_i = (1 - z_i)/2$, Eq. (1) becomes an Ising Hamiltonian

$$\tilde{H} = \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j, \quad C = s \tilde{H} + c, \quad (3)$$

where the identity term has been dropped and the coefficients rescaled so that $\max(|h|, |J|) = 1$. The rescaling matters in practice because the pattern-search step size and the shot-noise tolerance of Sec. V are expressed in these units, and without it they would have to be retuned for every instance. In these units the uniform superposition has $\langle \tilde{H} \rangle = 0$, so a run that ends at a positive value has done worse than the trivial state. Each notebook checks that the diagonal of Eq. (3) reproduces Eq. (1) on all 2^n bitstrings before any circuit is built.

The four instances are nested: instance n consists of the first n assets of a fixed twelve-asset table of large-capitalization US equities, with $B = n/2$. Any change in the results with n is therefore a property of the same problem growing rather than of four unrelated problems. Their parameters and exact solutions are listed in Table I.

We report three quantities. The *feasible probability mass* is the fraction of shots with Hamming weight B ; post-selecting on it is a free form of error mitigation, since every other bitstring is certainly an error. The *expected approximation ratio conditioned on feasibility* is

$$r = \frac{C_{\max}^{\text{feas}} - \langle C \rangle_{\text{feas}}}{C_{\max}^{\text{feas}} - C_{\min}}, \quad (4)$$

an average over the whole post-selected distribution, which, unlike a best-of- N figure, cannot be improved simply by taking more shots. The third is $p(\text{optimum} \mid \text{feasible})$, the probability of sampling the exact optimum among feasible outcomes, which we report as a multiple of the uniform value $1/\binom{n}{B}$.

The optimizer does not minimize Eq. (4) directly. It minimizes the conditional value at risk [27], CVaR $_{\alpha}$, the mean energy of the lowest α fraction of the sampled probability mass, with $\alpha = 0.5$. On a noisy device the mean

energy is dominated by the depolarized background while the low-energy tail still carries the signal; $\alpha = 0.5$ was the best of the three values tried in simulation (Appendix A); a shorter tail carries less of the background but more shot noise. Related tail-based objectives are discussed in Ref. [28].

III. CIRCUIT SYNTHESIS ON A CHAIN

The cost Hamiltonian of Eq. (3) couples every pair of assets, so its interaction graph is complete, while the processor’s coupling map is a heavy-hexagonal lattice. Handing the ansatz to a preset transpiler produces a routing solution whose depth is set by a SWAP heuristic and varies with the random seed, which is inconvenient when the circuit is meant to be the fixed, independent variable of an experiment.

We instead place the n qubits on a single physical chain and realize all $\binom{n}{2}$ couplings with an odd–even transposition network [5, 29, 30]: at each of n steps, disjoint neighbouring pairs interact and then swap, so that after n steps every pair has met exactly once. Writing $\text{RZZ}(\theta) \cdot \text{SWAP}$ explicitly as

$$\text{cx}(a, b) \text{ rz}(\theta, b) \text{ cx}(b, a) \text{ cx}(a, b), \quad (5)$$

which is exact up to a global phase, costs three entangling gates per pair. The resulting gate count is deterministic,

$$N_{2q} = 3p \binom{n}{2}, \quad D_{2q} = 3pn, \quad (6)$$

independent of the transpiler seed, and the final permutation of qubits is tracked so that classical bit i always carries logical qubit i .

Emitting Eq. (5) by hand rather than leaving it to the compiler is worth the small effort: Qiskit’s two-qubit block consolidation does not fuse gates that carry unbound parameters, so a transpiled $\text{RZZ} \cdot \text{SWAP}$ costs five entangling gates instead of three. Table II and Fig. 1 compare the hand-built network with the preset pass manager at optimization level 3 for the four circuits actually submitted; the network uses somewhat fewer two-qubit gates and is between two and three times shallower. The chain itself is chosen at run time by a beam search over simple paths of the coupling map that minimizes $\sum -\log(1 - \varepsilon)$ over the two-qubit and readout errors along the path, so that the circuit sits on the best line the device offers on the day, and the notebook checks that the pass manager left the layout in place. Each circuit is verified against the exact QAOA state vector before submission.

IV. WHAT A JOB COSTS

IBM defines the metered usage of a job as the time for which a QPU is locked to execute it [31]. The documentation gives an estimate of the form (*per-sub-job overhead*)

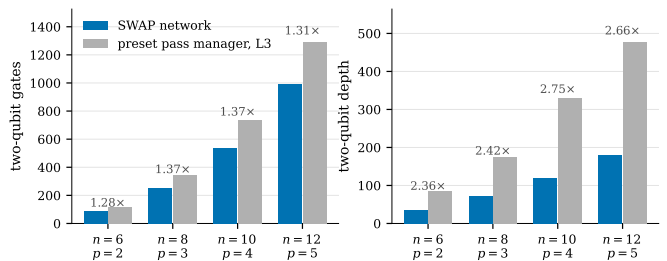


FIG. 1. Two-qubit gate count and two-qubit depth of the swap-network synthesis compared with Qiskit’s preset pass manager at optimization level 3, for the four circuits actually submitted. Labels give the ratio.

TABLE II. Circuit resources on `ibm_fez`. The swap-network columns follow Eq. (6); the preset columns depend on the transpiler seed.

n	p	swap network		preset, level 3		ratio	
		N_{2q}	D_{2q}	N_{2q}	D_{2q}	gates	depth
6	2	90	36	115	85	1.28	2.36
8	3	252	72	344	174	1.37	2.42
10	4	540	120	738	330	1.37	2.75
12	5	990	180	1293	478	1.31	2.66

+ ($\text{rep_delay} + \text{circuit length}$) $\times$ ($\text{number of executions}$), with an overhead of roughly two seconds per sub-job. Every job of this study submitted a single PUB in job mode — no `Session` and no `Batch` — which is the access mode available on the free Open Plan [32].

Instantiating that estimate for `ibm_fez`, with its 250 μs default repetition delay and the gate durations from its calibration data, and rounding the overhead up to three seconds for budgeting, gives

$$T_{\text{QPU}} \simeq 3 \text{ s} + (352.6 + 0.267 D_{2q}) \mu\text{s} \times N_{\text{shots}}, \quad (7)$$

with D_{2q} the two-qubit depth. The per-shot coefficient is dominated by the repetition delay and readout; the depth-dependent part is small for the circuits used here. Equation (7) is what the ledger described in Appendix B used to budget each run in advance.

We had also intended to record the runtime’s own reported usage for every job, and this did not work: the key path used to read `job.metrics()` does not exist in the client version we ran, and the surrounding `try/except` fell back to the predicted value without warning. We discovered this only when auditing the released data, in which the recorded and predicted usage are consequently identical. Every figure for metered QPU time in this paper is therefore Eq. (7) evaluated on the job’s contents, not a measurement of the device. We expect it to be close, since it follows IBM’s own estimator, but no conclusion below depends on it, and we say so wherever it is quoted.

What was measured for every job is the wall-clock time from submission to result. Across the 193 jobs of this

study the median round trip was about 19 s, the shortest about 11 s, and the distribution has a long tail with one job taking about 24 minutes (Fig. 2). There is, in other words, a floor of order ten seconds on a round trip that no reduction in the contents of a job removes, and over all 193 jobs the wall-clock time exceeded the estimated metered time by a factor of about four (by more for the four main runs alone; Table III). Table III lists the resources used by the four main runs; the depth scan and the replicate study added 25 and 66 jobs respectively.

The consequence for a variational loop is the same under either accounting. An optimizer that submits one job per function evaluation pays the per-job overhead once per evaluation, before any shot is taken; a batched iteration that puts M parameter vectors into one PUB pays it once for all M . At the settings used below M is between 10 and 22.

Two qualifications belong here. First, wall-clock time is not a property of the device or of the algorithm: it depends on other users' load and on fair-share priority, and our own per-run ratios of wall-clock to estimated metered time vary by a factor of about four between runs (Table III). It should be read as a characterization of the access mode on the days we used it, not as a constant. Second, that access mode is job mode on the free plan. IBM's **Batch** and **Session** modes exist precisely to amortize queueing across a sequence of jobs, and a user with access to them would see a smaller wall-clock penalty per job, though the per-job metered overhead of Eq. (7) is charged in all three modes [31]. The argument of this paper is therefore strongest for the open-plan user and weaker, but not absent, for one with reserved capacity.

V. BATCHED PATTERN SEARCH

Pattern search, also called compass search or direct search, replaces derivatives with a fixed stencil of trial points around an incumbent, accepting a move only when a trial point improves on the incumbent and contracting the stencil when none does [17, 18]. The property that matters here is structural rather than statistical: the entire iteration is determined by the incumbent and the current radius, so every point can be named before any value comes back. The classical direct-search literature exploited this parallelism long ago [19]; we simply apply it to a job queue.

Algorithm 1 states the variant used. For $2p$ parameters an iteration consists of the incumbent (row 0), the $2 \times 2p$ axial probes at radius ρ , and, when the previous iteration moved, one momentum probe continuing along the accepted step: at most $4p + 2$ points, all submitted as the rows of a single PUB in a single job.

Two details deserve comment. The acceptance test requires an improvement larger than ϵ , half the standard error of the measured energy of the incumbent row. Without it, at the shot counts used here, the incumbent moves on nearly every iteration and the search follows

shot noise rather than the landscape; the same measurement that supplies the incumbent value supplies σ , so the tolerance costs nothing extra. The momentum probe uses the fact that the batch has a spare row whenever the previous iteration moved, and extrapolates along the accepted direction; it is a cheap substitute for a line search.

Because the number of candidates and the shots per job are both fixed in advance, the metered cost of a run of J iterations can be estimated from Eq. (7) before the run starts. A serial optimizer does not have this property: COBYLA's evaluation count is data-dependent and has to be truncated at some chosen point.

The initial point is the linear ramp $\beta_k = \Delta t (1 - (k - \frac{1}{2})/p)$, $\gamma_k = -\Delta t k/p$ with $\Delta t = 0.5$, an adiabatically motivated schedule of the kind introduced in Ref. [2] and shown to avoid the poor local minima that random initialization often finds [33]. Warm-starting from a classical relaxation [34] and parameter transfer [35–37] are alternatives we did not use. The sign of γ follows the convention of Eq. (3).

VI. PROTOCOL AND CONTROLS

Each instance is run as a self-contained experiment with two arms under an identical budget (Fig. 3).

Matched budget. Both arms receive the same number of jobs J and the same shots per job S_{job} . The batched arm spends a job on $M = 4p + 2$ parameter vectors at S_{job}/M shots each; the serial arm spends a job on one parameter vector at the whole S_{job} . The two arms therefore consume the same quantum resources by both measures, and the difference under test is how a job's shots are allocated.

This constraint nominally gives the serial arm M times as many shots per evaluation, or a $\sqrt{M}$ -fold advantage in standard error. In practice the simulations of Appendix A found the final quality insensitive to shots per candidate over the range used here, so a serial evaluation at tens of thousands of shots is well into the regime where extra shots buy little. We keep the constraint because it removes shot count as an alternative explanation of the result, not because it is a large handicap for the batched arm.

One asymmetry runs the other way and should also be stated. The batched arm's M rows execute inside a single job under one calibration, so comparisons among them are free of inter-job drift; COBYLA's successive evaluations sit in different jobs, minutes apart, and carry that drift on top of shot noise. This favours the batched arm and is part of the difference under test, but it means that difference is not purely one of allocation.

The serial optimizer. COBYLA [11] with ρ_{beg} equal to the initial pattern-search radius. It was chosen because it was the strongest of COBYLA, SPSA, Nelder–Mead and Powell in the simulations of Appendix A at every size over the range of job budgets used here; the reference should be the best available serial method rather

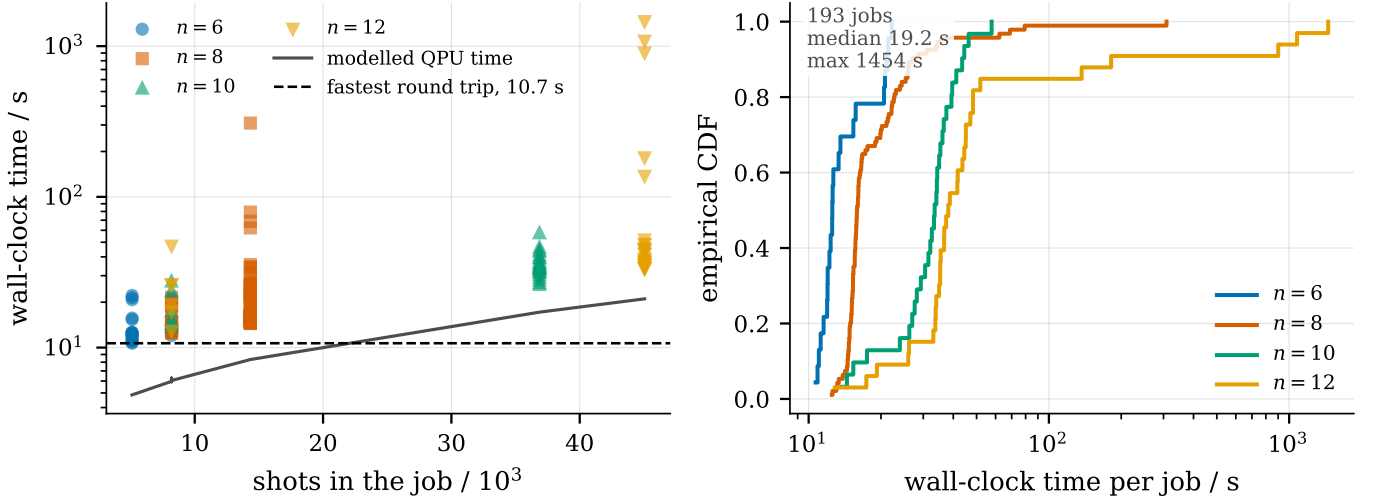


FIG. 2. The cost of a job on `ibm_fez`, from the 193 jobs of this study. **Left:** measured wall-clock time from submission to result against the number of shots the job carried. The dashed line marks the shortest round trip observed. **Right:** the distribution of the same times, by instance size. Both panels are measured; the estimated metered time of Eq. (7) is shown for scale only.

TABLE III. Resource ledger for the four main runs. “QPU” is Eq. (7) evaluated on each job’s contents, an estimate rather than a measurement for the reason given in the text; “wall” is the measured sum of submission-to-result times. The two follow-up studies added a further 91 jobs. Every job identifier is in the released data.

n	p	shots/cand.	shots/job	jobs	total shots	QPU / s	wall / s	ratio	median wall / s
6	2	512	5,120	20	111,600	100	292	2.9	13
8	3	1024	14,336	24	325,612	193	942	4.9	23
10	4	2048	36,864	28	946,160	448	968	2.2	34
12	5	2048	45,056	30	1,241,066	587	4688	8.0	38
total				102	2,624,438	1329	6890	5.2	

Algorithm 1. Batched pattern search. One iteration is one job.

Require: incumbent $\mathbf{x}_0$, radius ρ , shot budget S_{job} per job

- 1: $\mathbf{x} \leftarrow \mathbf{x}_0$, $\mathbf{x}_{\text{prev}} \leftarrow \mathbf{x}_0$
- 2: **for** $j = 1 \dots J$ **do** $\triangleright J = \text{the job budget}$
- 3: $\mathcal{C} \leftarrow \{\mathbf{x}\} \cup \{\mathbf{x} \pm \rho \mathbf{e}_i\}_{i=1}^{2p}$
- 4: **if** $\mathbf{x} \neq \mathbf{x}_{\text{prev}}$ **then**
- 5: $\mathcal{C} \leftarrow \mathcal{C} \cup \{2\mathbf{x} - \mathbf{x}_{\text{prev}}\}$ $\triangleright \text{momentum probe}$
- 6: **end if**
- 7: submit **one job**: one PUB, $|\mathcal{C}|$ parameter rows, $S_{\text{job}}/|\mathcal{C}|$ shots each
- 8: $v_c \leftarrow \text{CVaR}_\alpha$ of the distribution measured for $c \in \mathcal{C}$
- 9: $\sigma \leftarrow \text{standard deviation of } \hat{H} \text{ under row } 0$
- 10: $\epsilon \leftarrow \frac{1}{2} \sigma / \sqrt{S_{\text{job}}/|\mathcal{C}|}$ $\triangleright \text{shot-noise tolerance}$
- 11: $c^* \leftarrow \arg \min_c v_c$
- 12: **if** $c^* \neq \mathbf{x}$ **and** $v_{c^*} < v_{\mathbf{x}} - \epsilon$ **then**
- 13: $\mathbf{x}_{\text{prev}} \leftarrow \mathbf{x}$, $\mathbf{x} \leftarrow c^*$, $\rho \leftarrow \min(1.3\rho, \pi/2)$
- 14: **else**
- 15: $\rho \leftarrow \max(0.6\rho, 0.02)$
- 16: **end if**
- 17: **end for**

than a weak one. (SPSA is stronger at a budget of two jobs and Powell overtakes beyond about forty; neither

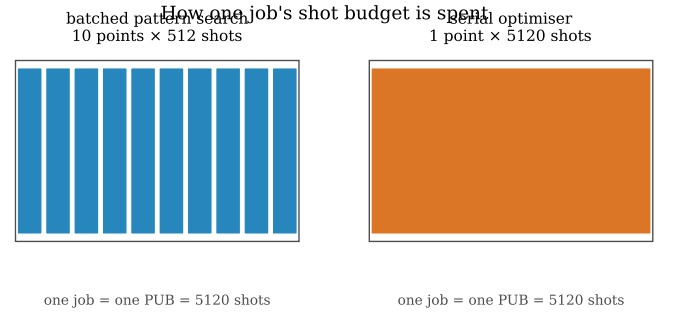


FIG. 3. How the two arms spend the shot budget of one job, drawn for the 6-asset settings. The batched arm divides the job’s shots among $4p + 2$ parameter vectors; the serial arm spends all of them on one.

is the regime studied.) This ordering is consistent with published comparisons of optimizers under noise [38, 39].

Error suppression. Dynamical decoupling [40, 41] on idle qubits and Pauli twirling of gates and measurements [42] were enabled for every job; neither consumes extra shots. No zero-noise extrapolation or probabilistic error cancellation [43] was used, so apart from these

two settings the measured distributions are raw device output, and the only mitigation applied anywhere in the paper is post-selection on Hamming weight.

The $\gamma = 0$ control. After the two optimization arms, one job is submitted at the batched arm’s best β with γ set to zero. The circuit has the same gate count and depth, but its ideal output is exactly $|+\rangle^{\otimes n}$. Any structure that survives in its output is the device’s noise floor rather than something the algorithm found, and every result below should be read against it.

The drift control. The two arms run one after the other, so a device that drifts over the half hour of a run would favour whichever arm ran under better conditions. The last job of every run repeats the first batched iteration exactly. The shift in measured CVaR between the two was of the order of the shot noise and of either sign at the four sizes, so no drift correction is applied.

Two views of quality. Because $n \leq 12$, the noiseless output distribution of any parameter vector can be computed exactly. Alongside what the device measured we therefore report the exact noiseless approximation ratio of the best point each arm holds after each job. For the batched arm “holds” means the incumbent, the point that has survived the acceptance test of Algorithm 1, and the curve is the running maximum of its noiseless ratio. For the serial arm it is the running maximum over all the points COBYLA has evaluated, an upper bound on what COBYLA would return. This metric is generous to the serial arm, since the batched arm is credited only with points its acceptance rule kept while COBYLA is credited with the best of everything it touched. It is also an oracle: neither optimizer can compute it, and it is unavailable at any size that could not be simulated. It is a diagnostic of the search rather than a quantity either method returns, and the device read-outs of Sec. VII are its non-oracle counterpart.

Hyperparameters. The objective (α), the initial radius and the penalty μ were chosen on a noise model of `ibm.fez` before any hardware time was spent, and the same simulations guided the choice of shots and jobs (Appendix A). The depth p at $n = 10$ and 12 was set by hand above the range simulated; the consequences of this are examined in Sec. VIIE.

VII. RESULTS

A. Convergence per job

Figure 4 shows the two arms at the four sizes and Table IV summarizes it. The upper row is the running maximum of the exact noiseless approximation ratio of each arm’s incumbent, as a function of the number of jobs spent; the lower row is what the device reported.

The pattern is the same at all four sizes. After its first job the batched search already holds parameters of a quality that COBYLA first reaches after several jobs, and measured by the number of jobs needed to reach 95%

of its own final value, the batched search takes one or two where COBYLA takes several more (Table IV). The number of jobs COBYLA needs to catch up grows along the sequence, but we are cautious about reading that as a size effect: the batch width $M = 4p + 2$ and COBYLA’s initial-model cost $2p + 1$ both grow with p , and p was raised together with n , so instance size and batch width are confounded in these runs. The statement the data support is the weaker one, that the ordering holds at every size tested and, at $n = 8$, at every one of four starting points (Sec. VIIF).

The lower row of Fig. 4 is consistent with the following mechanism. The serial traces are jagged, because COBYLA spends its early jobs building a linear model from single points and several of those points are worse than where it started. The batched traces are nearly monotone, because each job contains its own incumbent together with a stencil around it, and the acceptance test rejects moves that are not larger than shot noise. The batched search is not finding better optima — measured by the best point either arm reached, the two end within about 0.01 of each other at every size — it is finding comparable ones in fewer round trips.

B. What the device delivered

Figure 5 and Table V report the read-out at each arm’s returned parameters, at 8192 shots, against the $\gamma = 0$ control and the uniform baseline.

Three things can be read from the table. First, the device output is well separated from its own noise floor at every size: the $\gamma = 0$ control tracks the uniform baseline closely in every case, while the optimized circuits sit well above it, by a margin that narrows as n grows. Second, the batched arm’s read-out is better than the serial arm’s at $n = 8, 10$ and 12 and marginally worse at $n = 6$. At $n = 10$ and 12 this follows the noiseless quality of the returned parameters (last two columns of Table IV); at $n = 6$ and 8 the two parameter sets are noiselessly equivalent, so the measured difference there reflects device noise rather than a property of the optimizer, and the replicate study of Sec. VIIF is the appropriate place to look for a systematic difference. Third, the probability of sampling the exact optimum among feasible outcomes is enhanced over uniform by more than an order of magnitude at every size.

The narrowing of the separation from the baseline with n has two causes, and Sec. VIIE separates them: part of it is a consequence of having increased the circuit depth along with the instance size, and part of it is the instance size itself.

C. Rank correlation with the exact cost

Figure 6 plots, for every feasible portfolio, the probability the device assigned it against its exact cost.

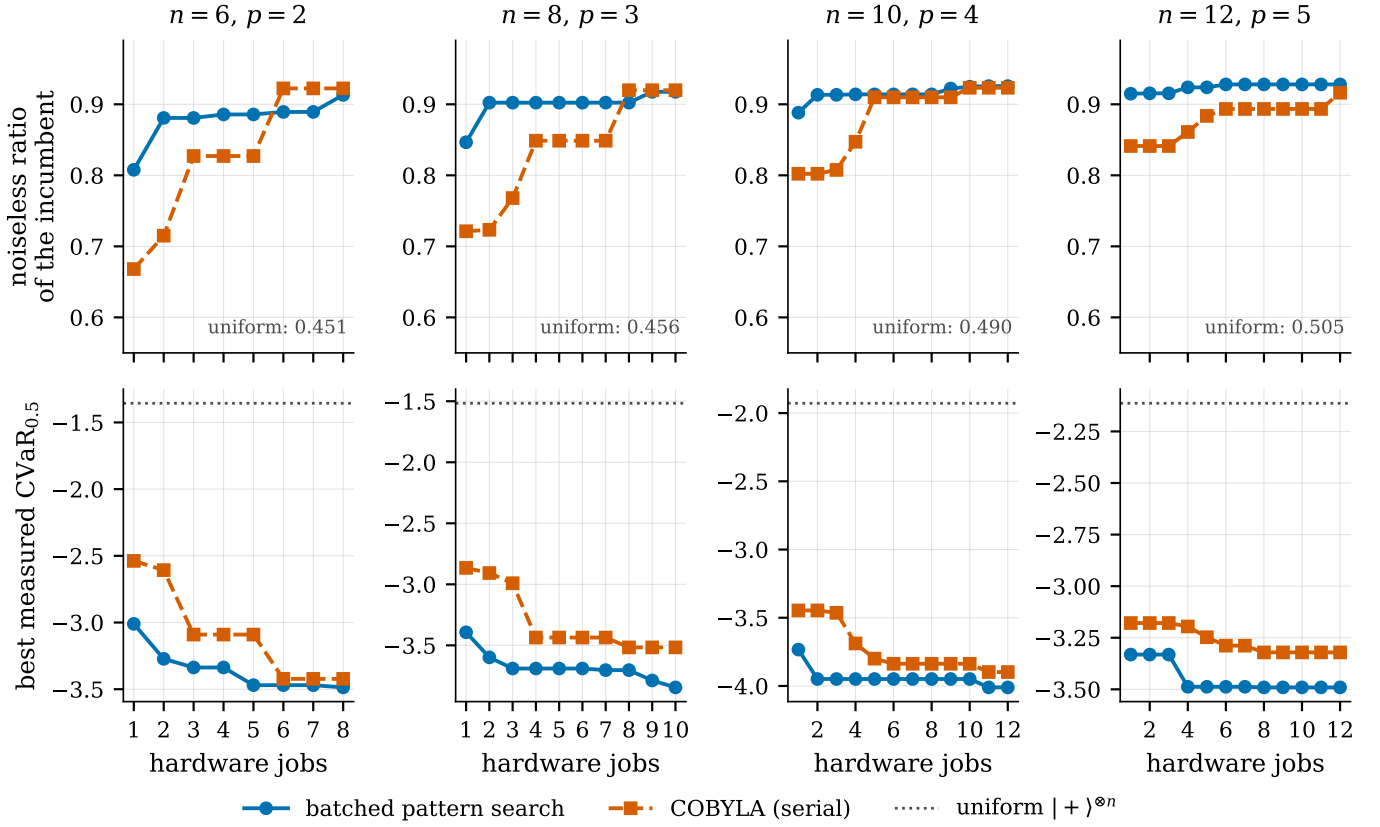


FIG. 4. Optimization on `ibm_fez` under a matched budget of jobs and shots. **Upper:** running maximum of the exact noiseless approximation ratio of the incumbent parameters, recomputed classically after every job; this is the quality of the point the optimizer has found, independent of how well the device can realize it. **Lower:** the best $\text{CVaR}_{0.5}$ measured on the device so far. Both arms receive the same number of jobs and the same shots per job; the batched arm spends a job on $4p + 2$ parameter vectors, the serial arm on one.

TABLE IV. Job efficiency on hardware. “COBYLA needs” is the number of jobs after which the serial arm first reaches the noiseless quality the batched arm had after its first job. “Jobs to 95%” is the number of jobs each arm needs to reach 95% of its own final noiseless ratio. “Best so far” is the running maximum at the end of the budget, and “returned” the noiseless ratio of the parameters each arm actually returned for the read-out of Table V. At $n = 12$ COBYLA matches the batched arm’s one-job quality only on the last job of its budget, and marginally.

n	p	jobs/arm	after one batched job		jobs to 95%		best so far		returned	
			r_{batched}	COBYLA needs	batched	COBYLA	batched	COBYLA	batched	COBYLA
6	2	8	0.808	3	2	6	0.913	0.922	0.913	0.922
8	3	10	0.847	4	2	8	0.918	0.920	0.918	0.920
10	4	12	0.888	5	1	5	0.926	0.923	0.926	0.883
12	5	12	0.915	12	1	5	0.928	0.916	0.923	0.891

The relationship is monotone, strong at the smaller sizes and weaker at $n = 12$ (Table V). The true optimum is the most probable feasible outcome at $n = 6, 8$ and 10 , and the second most probable of 924 candidates at $n = 12$, where the shot noise on a few tens of counts no longer resolves the leading few. The same statistic on the $\gamma = 0$ control is consistent with zero at every size, which is the check that makes the correlation meaningful: a circuit of the same depth on the same qubits, differing only in that

its cost-layer angles are zero, carries no information about the cost function. The control thus excludes any artefact of the readout chain or the gate set that is independent of the cost-layer angles.

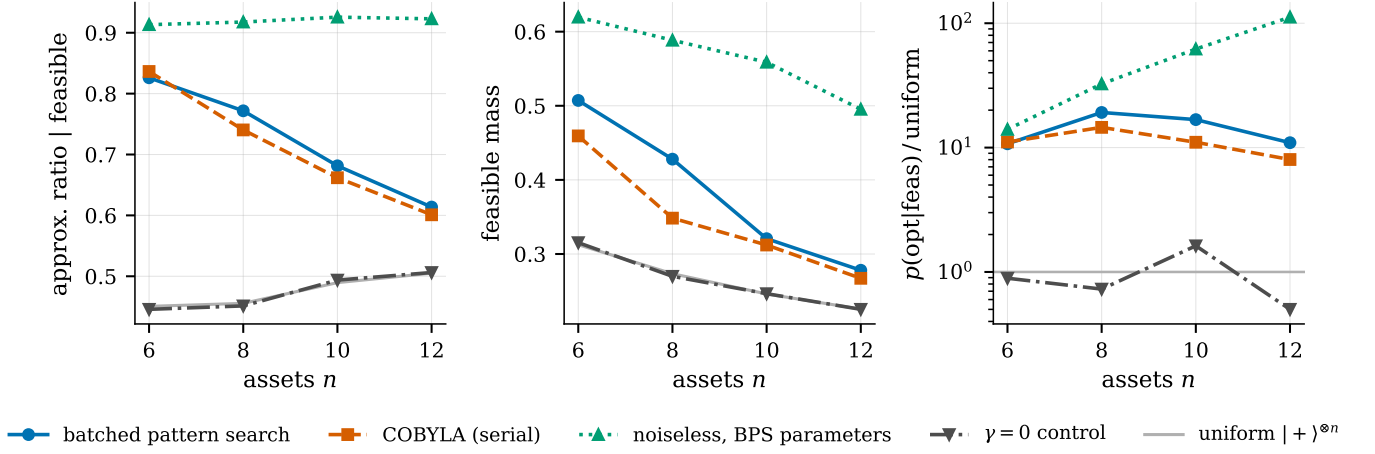


FIG. 5. Final read-out quality against instance size, at 8192 shots. The $\gamma = 0$ control runs a circuit of identical gate count and depth whose ideal output is $|+\rangle^{\otimes n}$; it tracks the uniform baseline at every size, which is what allows the separation between the measured curves and that baseline to be read as signal. The right panel is on a logarithmic scale; the downward arrow on the $\gamma = 0$ control at $n = 12$ marks an upper bound, that circuit having never sampled the optimum in 8192 shots.

TABLE V. Final read-out on `ibm_fez` at each arm's returned parameters. ρ_s is the Spearman rank correlation between the measured probability of a feasible portfolio and its exact cost; $\rho_s^{\gamma=0}$ is the same quantity for the noise-floor control.

n	approximation ratio feasible				feasible mass		$\frac{p(\text{opt} \text{feas})}{\text{uniform}}$	rank of opt.	rank corr.	
	batched	COBYLA	$\gamma=0$	uniform	batched	uniform			ρ_s	$\rho_s^{\gamma=0}$
6	0.826	0.836	0.446	0.451	0.507	0.312	10.7	1/20	-0.97	+0.07
8	0.772	0.740	0.451	0.456	0.428	0.273	19.2	1/70	-0.88	+0.07
10	0.681	0.662	0.494	0.490	0.321	0.246	16.8	1/252	-0.74	-0.08
12	0.614	0.601	0.506	0.505	0.278	0.226	11.0	2/924	-0.45	-0.00

D. The device as a depolarizing channel

The measured distributions are reasonably well described by a one-parameter model. Writing P_{ideal} for the exact QAOA distribution at the measured parameters, we fit

$$P_{\text{dev}} = \mathcal{R}[(1 - \eta)P_{\text{ideal}} + \eta\mathcal{U}], \quad (8)$$

where $\mathcal{U}$ is the uniform distribution and $\mathcal{R}$ applies independent readout bit flips at the calibrated rate, choosing η so that the model reproduces the measured mean energy: one fitted number per instance. The fitted η rises from about 0.25 at $n = 6$ to about 0.78 at $n = 12$ as the two-qubit gate count grows (Table II).

The four values are described by

$$\eta(N_{2q}) = 1 - \exp[-(0.19 + 1.35 \times 10^{-3} N_{2q})] \quad (9)$$

to within 0.03 (Fig. 7, left). The slope corresponds to an effective error of 1.35×10^{-3} per two-qubit gate, roughly half the median CZ error of 0.26% reported by the device calibration on the day. The two need not agree: a depolarizing channel acting on the output distribution is a weaker statement than one acting on the state, and Pauli twirling converts coherent errors into stochastic

ones, which are partly benign for a sampled distribution. The intercept is the depth-independent contribution from state preparation, single-qubit gates and measurement.

The right panel of Fig. 7 compares the model with quantities it was not fitted to. It reproduces the conditional approximation ratio and the feasible mass to within about 0.06 at every size, and under-predicts the loss in $p(\text{optimum} | \text{feasible})$, as one would expect: a global depolarizing channel damages the whole distribution uniformly, whereas correlated and coherent errors damage the sharp peak more. Equation (9) may therefore serve as a rough planning estimate for this device, converting a proposed (n, p) into an expected output quality, and is somewhat optimistic as a physical description. With four points and two parameters it should not be read as more than that.

E. Depth scan at classically optimal parameters

The circuit depth was increased along with the instance size in the runs above, which leaves the two effects entangled. They can be separated without an optimizer. For $n \leq 12$ the noiseless QAOA landscape can be optimized exactly and classically, so for each (n, p) we com-

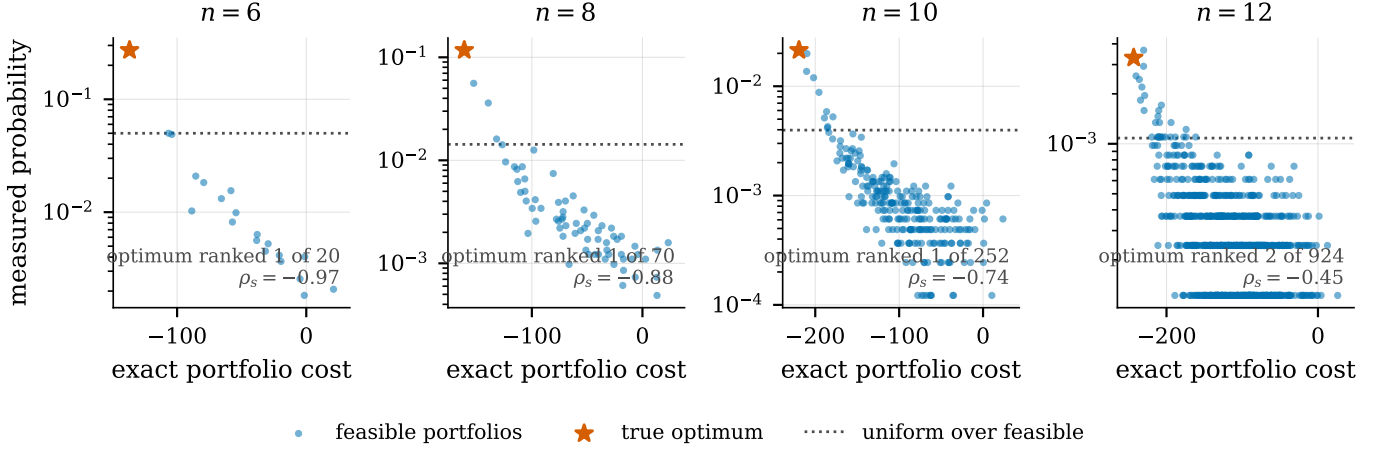


FIG. 6. Measured probability of every feasible portfolio against its exact cost, from the batched arm’s read-out. The star is the true optimum; ρ_s is the Spearman rank correlation. The corresponding correlation for the $\gamma = 0$ control is consistent with zero at every size (Table V).

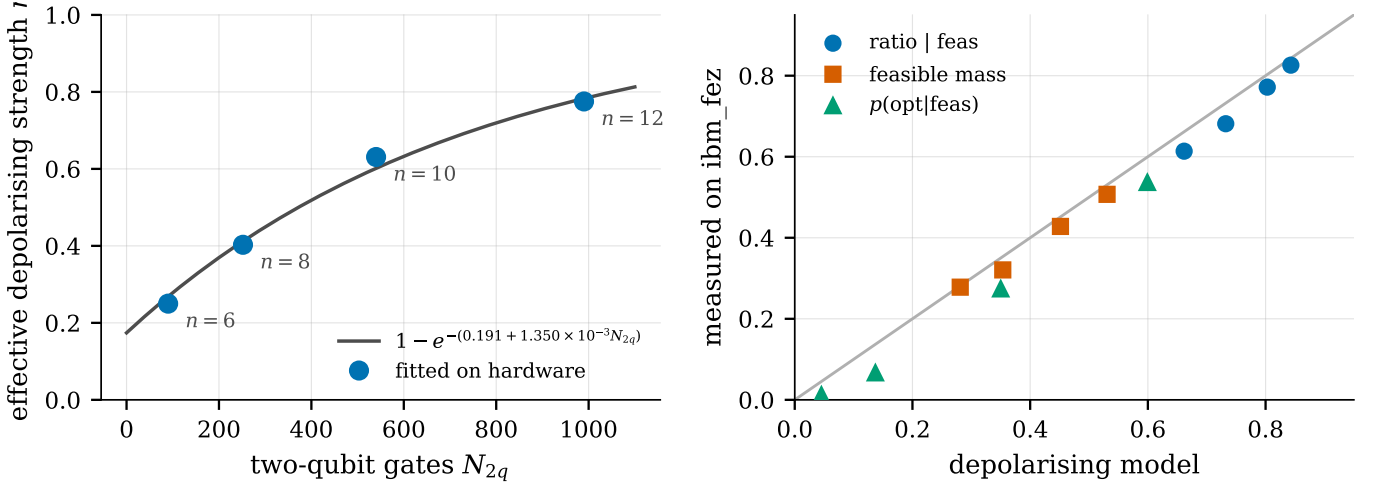


FIG. 7. **Left:** effective depolarizing strength η fitted to the measured mean energy of each instance, against the two-qubit gate count of its circuit; the line is Eq. (9), a two-parameter fit to the four points. **Right:** the model of Eq. (8), with η fitted only to the mean energy, against the measured values of three quantities it was not fitted to.

puted the best parameters any optimizer could return at that depth and then measured what the device made of them: one job of 8192 shots per (n, p) , for four sizes and five depths, plus a second parameter set at $n = 12$ described below. This removes the optimizer from the picture and leaves only the depth–noise trade-off.

Figure 8 and Table VI show the expected shape. The noiseless ratio rises with p at every size and exceeds 0.95 by $p = 5$. (At $n = 6$ it dips very slightly between $p = 3$ and 4, which is the multistart classical optimizer missing the global optimum of an eight-parameter landscape rather than anything physical.) The measured ratio rises steeply from $p = 1$, peaks, and then declines. The peak lies at $p = 2$ or 3 at every size; these are single 8192-shot jobs, and at $n = 6$ and 8 the top two depths differ by less than shot noise, so the data support the statement that the optimum is at two or three layers but not which

of the two. At $n = 12$ the measured ratio falls by about 0.08 between $p = 2$ and $p = 5$ while the noiseless value continues to rise.

This partly separates the two effects mixed in Sec. VII. Comparing the last two columns of Table VI, the $n = 6$ and 8 runs were at the device optimum while the $n = 10$ and 12 runs were past it. Part of the decline across Fig. 5 is thus a depth effect rather than a property of the instance, though not all of it: even at each size’s own best depth the measured ratio still falls with n .

The second measurement at $n = 12$ addresses a follow-up question. If the loss comes from running too deep, could an optimizer that knew about the noise recover it by choosing different parameters at the same depth? We reoptimized against the model of Eq. (8), at the η predicted by Eq. (9), and measured the result. The noise-aware parameters differ from the noiseless-optimal ones

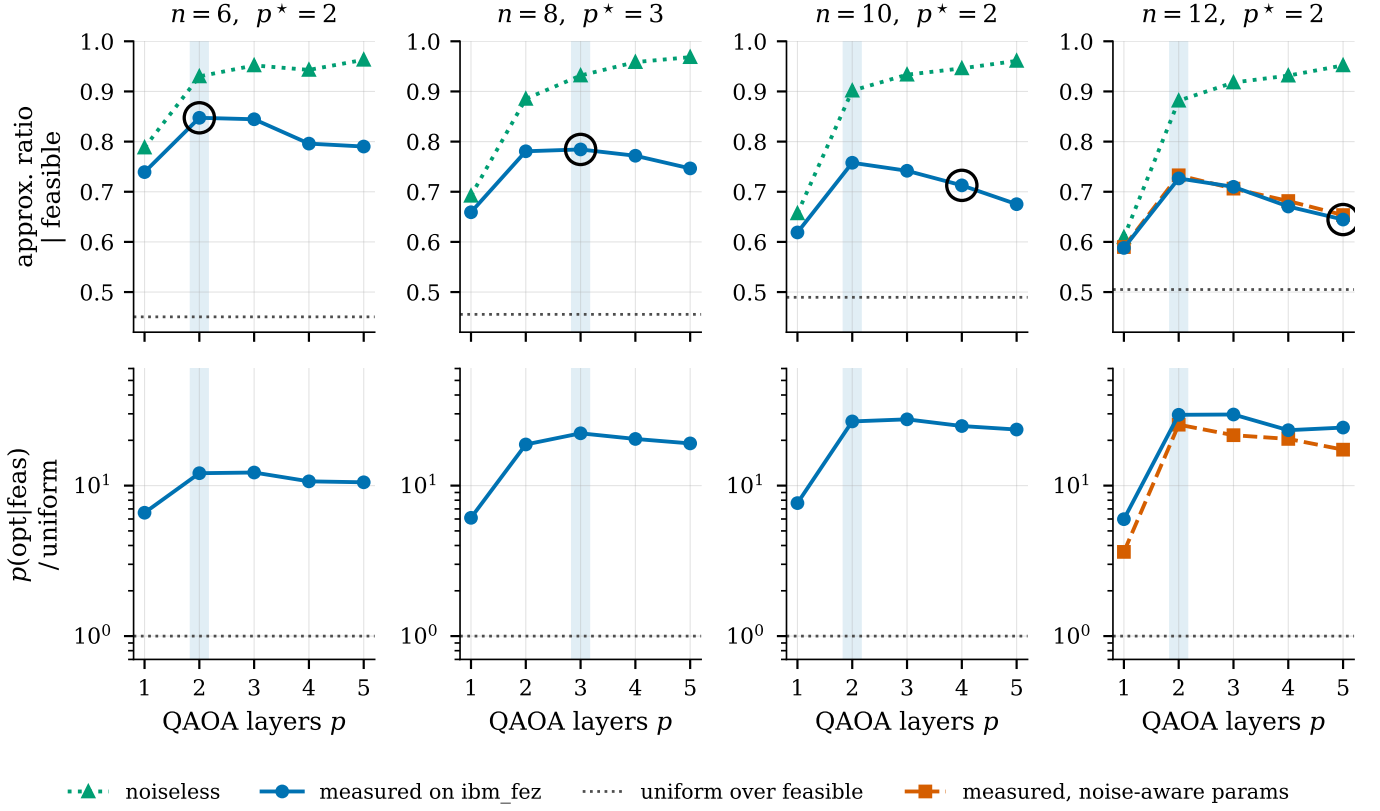


FIG. 8. Approximation ratio against circuit depth at classically optimal parameters. The noiseless curve (green) rises with p ; the measured curve (blue) rises, peaks at p^* (shaded), and falls, because the $3p\binom{n}{2}$ two-qubit gates eventually cost more fidelity than an extra layer gains. Black rings mark the depth actually used for that size in the main runs. At $n = 12$ the orange curve repeats the measurement at parameters optimized against the noise model of Eq. (8) instead of the noiseless landscape. **Lower:** the probability of sampling the exact optimum, conditioned on feasibility, as a multiple of the uniform value $1/\binom{n}{B}$.

by at most about 0.01 in measured ratio, in either direction, at every depth (orange curve, Fig. 8). At a given depth, in other words, reoptimizing against the noise model changed the measured ratio by about 0.01 at most. This is consistent with the picture in which the dominant error is a global depolarizing channel, which contracts the whole output distribution toward uniform and therefore cannot be steered around by a change of parameters; that noisy devices are limited in this way, and that the limitation tightens with circuit volume, has been established in general terms [44, 45], of which this is a small concrete instance.

One heuristic consistent with these data is to choose p by maximizing the product of the noiseless ratio and $1 - \eta(3p\binom{n}{2})$, with η from Eq. (9). For `ibm_fez` at these sizes it selects $p = 2$ at every size, in line with the measured optimum lying at two or three layers. It requires an estimate of η for the device in question.

F. Repeatability across initializations

The comparison above is one trajectory per arm per size. To attach a spread to it we repeated the 8-asset

experiment from three further starting points, changing nothing else: same instance, same $p = 3$, same shots, same ten jobs per arm. The starts are the linear ramp at $\Delta t \in \{0.35, 0.50, 0.60, 0.85\}$, with $\Delta t = 0.50$ the run already reported; within a start both arms receive the same initial point.

The ordering holds at every start (Fig. 9, Table VII): the batched search reaches 95% of its own final value in one or two jobs at each of the four starts, and COBYLA takes several more at each.

Two further features are worth noting. The final noiseless values of the two arms are statistically indistinguishable, with means that differ by less than the standard deviation of about 0.013 across starts, which is the same message as before: the two methods find equally good parameters and differ in how many round trips they take to do so. The device read-outs at those parameters (Fig. 9, right) show a small edge for the batched arm at each of the four starts, of the order of the spread, with both arms far above the $\gamma = 0$ floor. And on these four starts the advantage was largest at the poorest one, where COBYLA takes longest to converge, and smallest at the best, where the ramp already lands near a good region and both arms converge within a few jobs. If that holds more generally,

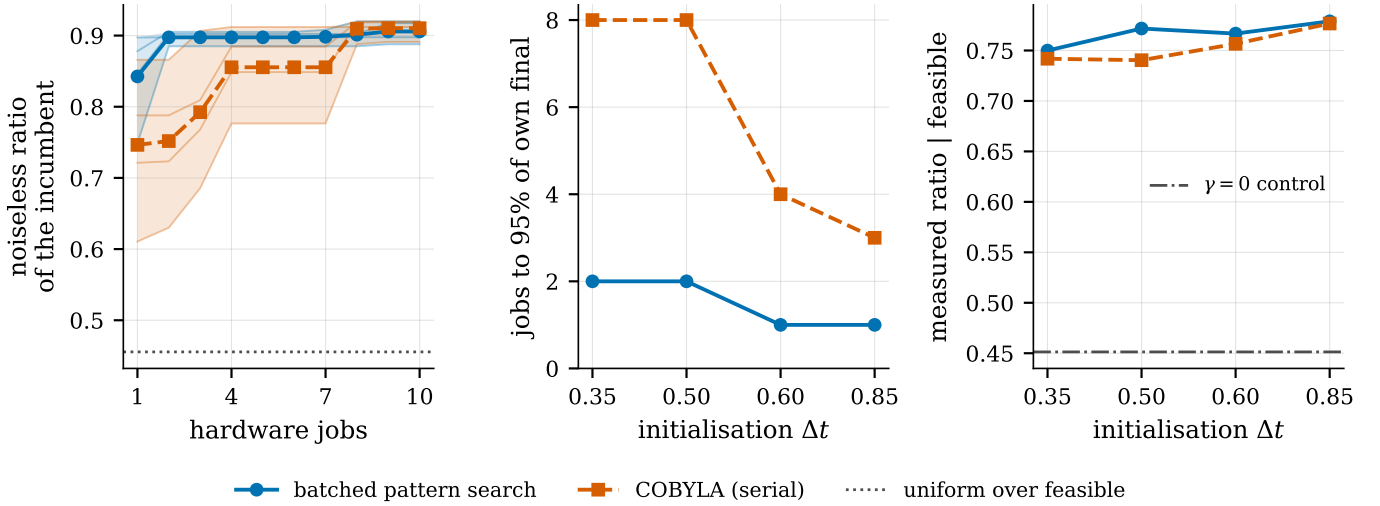


FIG. 9. Four independent initializations at $n = 8$, $p = 3$. **Left:** noiseless approximation ratio of the incumbent against jobs; thin lines are individual starts, markers the mean, bands the full range. **Centre:** jobs each arm needed to reach 95% of its own final value. **Right:** measured approximation ratio of the read-out at each arm’s returned parameters, against the $\gamma = 0$ noise floor.

TABLE VI. Depth scan on `ibm_fez`, 8192 shots per point, at classically optimal parameters. First row of each triple: measured approximation ratio. Second, in *italic*: the exact noiseless value at the same parameters. Third: the measured $p(\text{optimum} \mid \text{feasible})$ as a multiple of the uniform value. p^* is the depth with the highest measured ratio and “ p run” the depth used in the main runs.

n	$p = 1$	$p = 2$	$p = 3$	$p = 4$	$p = 5$	p^*	p run
6	0.739	0.847	0.845	0.796	0.790	2	2
	<i>0.788</i>	<i>0.930</i>	<i>0.952</i>	<i>0.943</i>	<i>0.963</i>		
	7×	12×	12×	11×	11×		
8	0.659	0.781	0.784	0.772	0.747	3	3
	<i>0.692</i>	<i>0.885</i>	<i>0.932</i>	<i>0.959</i>	<i>0.968</i>		
	6×	19×	22×	20×	19×		
10	0.619	0.758	0.742	0.713	0.675	2	4
	<i>0.657</i>	<i>0.902</i>	<i>0.933</i>	<i>0.946</i>	<i>0.961</i>		
	8×	27×	28×	25×	24×		
12	0.588	0.727	0.710	0.671	0.645	2	5
	<i>0.609</i>	<i>0.882</i>	<i>0.918</i>	<i>0.932</i>	<i>0.952</i>		
	6×	30×	30×	23×	24×		

a practitioner who cannot pre-tune the initialization — which is the situation whenever the instance is too large to simulate — is the one for whom the batched method would help most.

VIII. DISCUSSION

A. What the comparison does and does not show

The batched search does not find better optima than COBYLA in any reliable sense. Measured by the best point each arm reached, the two end within about 0.01 of each other at every size; measured by the parameters each arm actually returned, the two are level at $n = 6$ and 8 and the batched arm is ahead by a few hundredths at $n = 10$ and 12 (Table IV). What the batched search does is reach comparable parameters in fewer jobs, and on the access mode of Sec. IV jobs were the dominant cost. The claim is about the shape of the convergence curve near its origin, not about its asymptote, and the regime it speaks to is the one in which a user has time for a handful of round trips rather than a hundred.

Several limitations bound the strength of the evidence, and we list them here.

Replication. The four main instances were each run once. The replicate study of Sec. VIIF attaches a spread to the 8-asset comparison, but the 10- and 12-asset results are single trajectories, and the differences in final quality there should be read against the replicate standard deviation of about 0.013 rather than as precise.

Sequential arms. The two arms ran one after the other rather than interleaved. The drift control of Sec. VI is a single paired repeat per instance, which cannot separate drift from shot noise; it bounds the effect at a few hundredths of a CVaR unit and no more than that. Interleaving the arms would have been cleaner and would have cost nothing.

Ablations not run. Two further arms would sharpen the attribution and each would cost one job budget. A

TABLE VII. The 8-asset comparison from four initializations. “COBYLA matches at” is the job at which the serial arm first reaches the noiseless quality the batched arm had after its first job; “best so far” is the running maximum of the noiseless ratio at the end of the budget and “read-out” the measured approximation ratio at the returned parameters. [†] marks the run reported above.

Δt	after one job		COBYLA	jobs to 95%		best so far		read-out	
	batched	COB	matches at	batched	COB	batched	COB	batched	COB
0.35	0.748	0.611	4	2	8	0.888	0.891	0.750	0.742
0.50 [†]	0.847	0.721	4	2	8	0.918	0.920	0.772	0.740
0.60	0.878	0.788	4	1	4	0.920	0.919	0.767	0.757
0.85	0.897	0.866	3	1	3	0.898	0.912	0.779	0.777
mean	0.843	0.746		1.50	5.75	0.906	0.910	0.767	0.754

random-batch arm, with M points per job drawn around the same start and tracked greedily, would separate the value of batching itself from the value of the pattern-search stencil. A serial pattern search, one point per job with the same acceptance rule, would separate the stencil from the shot-noise tolerance of Algorithm 1, which is a noise-robustness feature COBYLA has no counterpart of. As designed, the experiment measures the package rather than its parts.

A batchable competitor. COBYLA was chosen because it was the strongest serial method in the simulations, but as noted in Sec. I, SPSA and parameter-shift gradients are themselves batchable, as are model-based trust-region methods [46], multistart schemes [47] and information-sharing schemes [48]. A batched SPSA, or several concurrent COBYLA chains packed into one PUB, would be the harder comparison, and we did not run it.

The oracle metric. The per-job curves of Fig. 4 use the exact noiseless ratio, which no optimizer can compute and which does not exist at a problem size worth solving. It diagnoses the search; it is not a quantity either method returns. The device read-outs of Tables V and VII are the non-oracle counterpart, and there the two arms are closer.

Selection bias in the measured traces. The lower row of Fig. 4 plots a running minimum of measured CVaR. The batched arm takes that minimum over M values per job and the serial arm over one, so the batched trace carries a downward selection bias. This is one reason the separation is larger there than in the independently measured read-outs, and the read-outs are the numbers to rely on.

Problem size. Twelve qubits is far below the size at which any quantum method could matter for portfolio selection, and the instances here are solved by enumeration in milliseconds. The sizes were chosen so that exact answers were available for every check in the paper. Nothing here bears on whether QAOA can compete with classical solvers at larger sizes.

B. Depth and instance size were varied together

The circuit depth was increased with the instance size, so the decline in measured quality across Fig. 5 mixes a depth effect with a size effect, and the depth scan of Sec. VII E only partly separates them. Repeating the $n = 12$ optimization at $p = 2$ rather than $p = 5$ is the obvious next experiment; Table VI suggests it would raise the final read-out, and it would not change the job-efficiency comparison, which is a statement about the optimizer at a fixed circuit.

C. Relevance beyond this experiment

The cost structure of Sec. IV is a property of the access mode rather than of QAOA. Any variational procedure driven through a queue — an eigensolver for chemistry [49], a quantum kernel evaluation, a calibration or error-mitigation sweep — pays the same fixed charge per round trip and can, in principle, fill the same PUB. Three suggestions follow.

First, when comparing optimizers for use on cloud hardware, report convergence against submitted jobs as well as against function evaluations. The two plots can look quite different, as Fig. 4 shows, and the job axis is the one a user can translate into time and cost. Application-oriented benchmarking efforts [50] that already report execution time including cloud overheads are a natural place for such an axis.

Second, among optimizers of comparable sample efficiency, prefer those whose next batch of query points is known in advance. The property admits degrees — pattern search names $4p + 2$ points per round trip, parameter-shift gradients $4p$, SPSA and its quantum-natural variant [51] two, and COBYLA, Nelder–Mead and Powell one — and an optimizer that is slightly worse per evaluation but names many points at a time can be considerably better per job.

Third, state the cost model alongside the result. Shots are reported because they were once the binding constraint; on the access mode used here they were not, and a paper that reports only shots leaves the reader unable

to reconstruct what the experiment would cost them.

We should also be clear about what a job-budget framing does not do. Nothing here addresses the fundamental obstacles to variational scaling, barren plateaus [52] and their noise-induced counterpart [45], and the narrowing separation between the measured curves and the noise floor in Fig. 5 is a reminder that at twelve qubits and nearly a thousand two-qubit gates most of the output distribution is already uniform. Job efficiency affects how quickly the device’s output quality is reached, not that quality.

IX. CONCLUSION

We compared a batched pattern search with COBYLA for optimizing QAOA parameters on IBM’s `ibm_fez` processor, with the two methods given the same number of submitted jobs and the same number of shots, on four nested portfolio-selection instances small enough to be solved exactly. The batched search, which evaluates its whole iteration in one job, reached a good parameter region after a single job at every size, where COBYLA needed several; the two methods converged to comparable final quality, and the ordering was unchanged across four starting points at the 8-asset size. The device’s output at the returned parameters was correlated with the exact ranking of portfolios and clearly separated from a control circuit of the same depth with the cost layer removed. A depth scan at classically optimal parameters located the device’s optimal circuit depth at two or three layers, beyond which additional layers reduced the measured quality, and reoptimizing the parameters against a noise model did not recover the loss.

The optimizer is deliberately simple. The observation we would draw from the exercise is that, when quantum processors are reached through a queue, the number of times an algorithm goes around the loop is a cost worth counting, and that counting it can change the choice of classical optimizer. We have tried to describe the experiment in enough detail, and to release enough of its material, that others can repeat it or improve on it.

ACKNOWLEDGMENTS

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DATA AND CODE AVAILABILITY

Every hardware result in this paper is produced by six self-contained Jupyter notebooks, one per instance size plus the replicate and depth-scan studies, which are released together with the raw data they produced: the complete job log including every IBM job identifier, the measured bitstring counts of every parameter row of every job, both optimization traces, the control read-outs and the final metrics. The figures and tables are generated from those files by the included scripts. The archive is available at <https://github.com/muf148/qaoa-job-budget>; code is released under the MIT licence and data under CC-BY-4.0.

Appendix A: Choosing the hyperparameters in simulation

Every hyperparameter was fixed on a noise model of `ibm_fez` before any hardware time was used. Full noisy simulation is too slow to sweep — one 12-qubit, $p = 2$ circuit at 2048 shots takes over a minute with a full Aer noise model — so the sweep used the one-parameter surrogate of Eq. (8), with η fitted for each (n, p) against Aer at two parameter points. The surrogate reproduced Aer’s CVaR to about 0.1 in normalized units and its feasible mass to about 0.05.

The sweep covered $p \in \{1, 2, 3\}$, shots per candidate $\in \{512, 1024, 2048\}$, jobs per arm $\in \{8, 12, 16\}$, $\alpha \in \{0.1, 0.25, 0.5\}$ and initial radius $\in \{0.25, 0.4\}$, restricted to configurations fitting a 300 s estimated budget, at ten random seeds each, followed by a confirmation study of the selected settings at forty seeds. It selected $\alpha = 0.5$ at every size and an initial radius of 0.25, and found the final quality insensitive to shots per candidate over the range swept and insensitive to the number of jobs beyond about ten. The shots and jobs actually used (Table III) were then set by hand within this guidance and within the hardware time available; the 300 s cap applied to the sweep, not to the final runs. On the surrogate the batched search was ahead of COBYLA at every job budget swept, with COBYLA needing sixteen jobs to match what the batched search had after twelve at $n = 6, 8$ and 10, and not matching it within sixty-four jobs at $n = 12$. The hardware runs show the same ordering under the different metric of Table IV, though the surrogate carries neither crosstalk nor drift and its absolute numbers should not be compared with the device’s.

The surrogate selected $p = 2$ at all four sizes, on the grounds that $p = 3$ improved the predicted ratio only slightly for 50% more two-qubit gates. The depths actually run were $p = 2, 3, 4, 5$, chosen by hand in the belief that a larger instance warrants a deeper circuit. The hardware depth scan of Sec. VIII E was consistent with the surrogate’s recommendation: the measured optimum is at two or three layers at every size. A fake backend carries no crosstalk and no drift, which is why the $\gamma = 0$

and drift controls of Sec. VI are part of every hardware run; but on the one question where its recommendation was overruled, it should have been followed.

Appendix B: Reproducing the experiment

Each notebook is self-contained and runs top to bottom against `ibm_fez` in roughly 100 to 600s of estimated metered time. A `DRY_RUN` switch replaces the device with the surrogate of Eq. (8) and executes the entire experiment in about a minute at no cost, which is the recommended first step. A ledger is enforced at run time: before every submission the notebook predicts the metered cost from Eq. (7) and refuses to submit a job that would exceed the configured cap.

Four implementation details caused errors that were silent rather than loud, and we record them because they would cost another user a run.

1. `BitArray.get_int_counts()` called with no argument merges the counts of all parameter rows of a PUB into one dictionary. A batched run that does this produces plausible-looking output that is

meaningless. The row index is mandatory.

2. The parameters of a transpiled circuit are ordered by name, not by construction order. The array of parameter values must be assembled by name lookup.
3. An explicit `initial_layout` must be checked to have survived the pass manager; otherwise the circuit can silently move off the chosen chain.
4. Reading the runtime's reported usage through `job.metrics()` is version-dependent, and a `try/except` around it will hide a wrong key path behind a plausible fallback. Ours did, on every job of this study, which is why the metered times reported in Sec. IV are estimates rather than measurements. If the device's own number is wanted, the code should check that it actually received one.

The first and the last of these have the same shape, a fallback that produces output indistinguishable from success, and we would encourage anyone building this kind of pipeline to make both fail loudly.

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